



WORLD CUP 2026

LIVE DASHBOARD

User Guide & Developer Deep Dive

A single-file, real-time tournament dashboard
Powered by the ESPN public feed · AI features by Claude

Edition 1.2

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CHAPTER 1

Welcome

The World Cup 2026 Live Dashboard is a single web page that follows the entire tournament in real time — live scores and a play-by-play match tracker, group standings, the full knockout bracket, the Golden Boot race, news, and AI match predictions. It runs entirely in your browser, installs to your phone's home screen like an app, and is hosted for free on GitHub Pages.

What makes it different

- **Always live.** Scores, stats, standings, lineups and ball-by-ball commentary stream from ESPN's public feed and need no account or key.
- **One file.** The whole application is a single `index.html` — nothing to install, no server, no database.
- **Yours to theme.** Pick a favorite nation and the entire app re-skins to that team's colors and follows them.
- **An immersive live tracker.** A ticking match clock, an animated pitch that follows the flow of play, live win probability, a momentum graph, commentary, lineups, and a goal celebration when your team scores.
- **Built-in AI.** Match predictions, player bios, the news fallback, and in-app goal clips are powered by Claude through a secure proxy — no key or setup for you (Chapter 8).

NOTE — Nothing to set up

Every core feature — scores, the live tracker, standings, the bracket, the Golden Boot, news and commentary — works with no AI at all. The AI extras (predictions, bios, clips) are powered through a built-in proxy, so there is still no key to enter (Chapter 8).

CHAPTER 2

The Interface at a Glance

The top bar

Across the top you'll find the app title, three controls, and the tab row:

- **Favorite pull-down (“★ Favorite”).** Choose any nation to make it your team — see Chapter 3. Once set, a **“Following · [team]” flag badge** appears in the header so your team is always in view.
- **Auto toggle.** Turns live auto-refresh on or off.
- **Alerts (bell).** Turns on goal notifications. The bell shows filled when alerts are on and tells you what happened when you tap it.

The seven tabs

Matches	Browse fixtures by day, see live/finished/upcoming games, and open any match. Your favorite team's next-match countdown sits at the top.
Standings	Live group tables for all twelve groups, plus a Road to the Round of 32 scenario card under each group (Chapter 6).
Bracket	The full Round-of-32-to-Final knockout tree, with your team's projected Path to the Final on top (Chapter 6).
Golden Boot	The top scorers. Tap a player to see each of their goals and watch a clip of each (Chapter 5).
News	Latest tournament headlines.
Predict	AI win/draw/win projections, predicted scorelines, and key factors for any matchup.
Fun	A reel of fun YouTube clips with filters including a Most Viewed worldwide cut (Chapter 7).

TIP — Tap anywhere on a match card

The whole card is a button. Live games open the Live Tracker, finished games open the review with highlights, and upcoming games open a data-rich preview (Chapter 4).

The live goal ticker

Whenever at least one match is in play, a **live ticker** slides in just beneath the tabs and stays visible on every tab. It shows every in-progress score as a tappable chip with the live minute, and a scrolling feed of **every goal across the whole tournament** — scorer, minute, and penalty/own-goal flags. The bar flashes and reads “GOAL!” the instant a new one lands anywhere. Hover or hold to pause the scroll; tap any score or goal to open that match. It disappears when no games are live.

CHAPTER 3

Your Favorite Team

Choosing one team ties three things together: the app's color theme, which team you “follow,” and the next-match countdown.

Two ways to pick

- **The header pull-down** labeled “★ Favorite.” Choose any nation; a “Following” flag badge then appears in the header.
- **On a team's page**, tap “**Make [team] my team.**” Open a team page by tapping its name or flag anywhere in the app.

What happens when you pick a team

- The whole app **re-themes** to that nation's colors.
- The team is **followed** (its star fills in and it appears in the “Following only” filter).
- A **countdown** to that team's next kickoff appears at the top of the Matches tab — days, hours, minutes, seconds — and switches to a LIVE badge when they're playing.

NOTE — Following extra teams

The star on each match row lets you follow additional teams for the “Following only” filter, without changing your themed favorite. Your one themed team is whatever you pick in the pull-down or via “Make my team.”

The team page

Tap any nation's name or flag to open its page. Alongside the group stats, form and next match, it includes:

- **Star players** — the team's scorers so far, each one a tap away from a full AI bio.
- **Full squad** — the complete roster, assembled from the team's recent match line-ups; tap any player to load their bio (scorers carry a goal count).
- **Recent results that open the full summary** — every recent result is tappable and jumps straight to that match's complete review (recap, goals, stats).

And once your team reaches a projected top-two slot, its **Path to the Final** appears at the top of the Bracket tab (Chapter 6).

HONEST LIMIT — Colors stay readable automatically

Some nations have very dark secondary colors (the USA's is navy). The app automatically lightens a too-dark accent so headings and labels never disappear against the dark background — so every team is a usable theme.

CHAPTER 4

Following a Match Live

Open any live game (the bright “Open Live Tracker” button) and you get a live-blog-style view designed to keep you in the match even when you can't watch the broadcast. It refreshes about every five seconds and updates incrementally as play unfolds.

Before kickoff: the match preview

Opening an **upcoming** match shows a data-rich preview built entirely from the live feed. For each side you get its group position and points, full W-D-L record, goals for and against and goal difference, the team's **key scorer** this tournament, and a **detailed recent-results list** — each result with the opponent (tappable) and the score. Below that sits any head-to-head from this tournament, and an opt-in, just-for-fun AI prediction. A **finished** match opens the same form guide alongside the official recap and the goal highlights.

During the match

Ticking clock	A minute-and-seconds clock that counts up between updates and re-syncs to the official minute. Handles half-time and stoppage time.
Big score	Pops gold the instant it changes.
Match progress bar	A marker sweeps from kickoff through half-time to full-time.
Animated pitch	A green field where a white ball follows the flow of play and a glow leans toward whichever side is pressing (see the note below).
Live highlights	The latest goals — each with a tappable clip — sit directly below the pitch and appear as goals go in.
Live win probability	A home / draw / away bar that swings with the score, time remaining and momentum. Clearly labeled as a model, not an official number.
Live stat tiles	Possession, shots, on-target, corners, fouls and cards — each flashes when its number changes.
Momentum + sparkline	A momentum bar plus a line graph that builds across the match.

Goal moments	Each goal with scorer and minute, plus an in-app highlight clip.
Live commentary	Real ball-by-ball text from the feed, newest first.
Lineups	Tap to expand both starting XIs, formations and substitutes.
Prediction	An AI read on the match — built in, opt-in.

Goal celebration

When the score changes while you're watching, a full-screen “GOAL!” takeover fires with confetti and the scorer's name, your phone vibrates, and — if you've switched on the horn (the speaker icon in the tracker header) — a short goal horn plays.

HONEST LIMIT — About the pitch and the ball

No free data feed provides the real ball's position — that comes from expensive optical tracking systems. The moving ball here is an **approximation driven by the live commentary and possession**: it heads toward a goal on a shot, to the corner flag on a corner, into midfield on a foul, and drifts into the dominant team's half. It tracks *where the action is*, not the literal ball.

TIP — Keep an eye on the other games

While you follow one match, the live goal ticker beneath the tabs (Chapter 2) keeps you across every other score and goal in the tournament — invaluable on the final, simultaneous group matchday.

CHAPTER 5

Goals, Highlights & the Golden Boot

The Golden Boot tab ranks the tournament's scorers — each with their **photo**, flag and a goal-race bar. Tap any player to open their card, which lists **every goal they've scored** — the minute, the match, the score, and penalty/own-goal flags — pulled straight from the match feed.

Comprehensive player profile

Opening a player now **auto-loads a full profile** (no extra tap): current club, age, position, preferred foot, height and squad number appear as chips, with dedicated sections for playing style, career arc, World Cup role and major honours. It is sourced from live web search and instructed never to invent facts, so any field it can't verify is simply left out. Profiles are generated by Claude through the built-in proxy (Chapter 8) and are **pre-loaded in the background** for the teams bound for the knockouts, so a tap usually opens instantly with no spinner.

Watching a goal

Each goal (in a player's card and in any finished match's review) has a **“Watch goal”** button. Tapping it embeds the official highlight **inside the app**, cued to that goal and trimmed to roughly ten seconds.

HONEST LIMIT — What the clips are — and aren't

These are the **official highlight videos embedded via YouTube**, automatically cued and trimmed to the goal. They are not rights-cleared, isolated broadcast cuts — that footage isn't available to a free app, and the app won't fake it. Resolving the right video uses AI web search (handled by the built-in proxy) and depends on the official clip existing on YouTube. Most goals resolve cleanly; some show an honest “no official clip found” state.

CHAPTER 6

Standings, Bracket, News & Predictions

Standings

Every group's table is computed live from finished results — points, wins, draws, losses, goals for/against and goal difference — with host nations flagged. Nothing is hand-entered; it all derives from the same match feed.

Road to the Round of 32

Under each group sits a scenario card that answers the question every fan asks late in the group stage: **who's through, who's out, and what does my team need?** The engine brute-forces every possible combination of that group's remaining results and labels each team **Through** (already top two), **3rd race+** (guaranteed at least third), **In the hunt** (still alive), or **Out** — each with a plain-English line such as “Win your next game → into the Round of 32” or “A draw is enough for a top-two finish.” Your favorite team is starred in its group.

HONEST LIMIT — It is an honest projection

Top two advance automatically, but the eight best third-placed teams across all twelve groups also qualify — so a “3rd race+” team is not mathematically safe until the group stage ends. The points maths is exact; goal-difference tiebreakers are simplified. Treat it as a very good projection, not a guarantee.

Bracket

The bracket shows the knockout path from the Round of 32 to the Final. During the group stage it is explicitly a projection: each slot shows the current projected qualifier and only **locks in a real team once that group is mathematically decided**. As knockout games finish, winners advance automatically.

NOTE — Group games aren't in the bracket

The bracket is knockout-only. A group-stage fixture (say, your team's next match) lives on the Matches tab and on the team's page, not in the bracket.

Path to the Final

When you've chosen a favorite, the top of the Bracket tab shows its **projected route to the trophy** — Round of 32, Round of 16, quarterfinal, semifinal, Final — with the likely opponent at each step. Opponents are the current projected qualifiers (the stronger seed advances on points, then goal difference), and each step turns green or red once a real knockout result is in. It assumes your team keeps winning: a dream-the-run visual, not a

forecast. Teams still in the best-third race appear once the group stage assigns their slot.

News

Headlines come from ESPN's feed; if that's unavailable, the app falls back to an AI-summarized roundup through the proxy.

Predict

Pick two teams (or use the upcoming-fixture shortcuts) for an AI projection: win/draw/win percentages, a predicted scoreline, and the key factors behind it. The same engine is built into every match modal.

IMPORTANT — Predictions are for fun

The win probabilities and predictions are model estimates, not betting advice.

CHAPTER 7

The Fun Tab

The Fun tab is a light-hearted reel built to keep you in the app between matches: short, good-natured videos of visitors experiencing America and of the tournament's most colourful travelling supporters.

What's in it

- **First-time-in-America reactions** — culture shock over giant portions, free refills, drive-thrus and Southern friendliness.
- **Flamboyant fans** — loud, vocal travelling supporters (Scotland's Tartan Army, Australia and more) taking over US cities.
- **Georgia & the South** — visitors discovering Atlanta, Waffle House, Buc-ee's and Peach State hospitality.

Filters

Chip filters across the top narrow the reel: **All, Most Viewed, World Cup, Food, Fans, Georgia, Atlanta, Shorts. Most Viewed** surfaces the hottest, most-viewed and most viral World Cup clips on the internet — it auto-loads when you tap the chip, and a **Most-viewed now** button refreshes it on demand. **Find fresh picks (AI)** pulls current, popular, embeddable clips via web search — including the latest most-viewed videos — and merges them in (de-duplicated); this one uses AI web search through the built-in proxy (Chapter 8).

Watching a clip

Tap any thumbnail and the clip opens in an **expanded player** over the page. Close it (the ×, tapping outside, or Esc) to return to the grid — only one clip plays at a time.

TIP — If a clip is silent, tap for sound

Phone browsers force the first unmute to come from a tap, so a clip may start muted. A **“Tap for sound”** button appears on the player — one tap enables audio. On desktop it plays with sound straight away.

HONEST LIMIT — Curated snapshot, not a guarantee

Picks reflect what's popular at curation time; rankings shift and creators can remove a video. Clips whose owners block off-YouTube playback show a clean “Watch on YouTube” panel instead. Everything here is kept good-natured and safe-for-work.

CHAPTER 8

AI Features & the Proxy

The AI extras — match predictions, player bios, the news fallback, in-app goal-clip lookups, and the Fun-tab “fresh picks” — are powered by Claude (Anthropic). In this deployment they run through a small **Cloudflare Worker proxy** that holds the API key on the server side, so **there is nothing for you to enter**. The core experience — scores, the live tracker, standings, the bracket, the Golden Boot, commentary — uses no AI at all.

Part A — For everyone: nothing to set up

There is no API-key field and no Settings gear in this build. The AI features simply work, because the owner has supplied the credentials behind the scenes through the proxy. Open the app and use it — predictions, bios, and clip lookups are already wired up.

NOTE — Why there's no key box anymore

Earlier versions asked each person to paste their own Anthropic key in Settings. That field is hidden whenever the app is served through the shared proxy — which is how this build is deployed — so friends and family never deal with keys, billing, or setup.

Part B — Self-hosting your own copy

If you fork and re-deploy the dashboard, you set the proxy up once. The key lives in the Worker, never in the page that visitors download.

1. Create an Anthropic key with a spend cap. At `console.anthropic.com`, open **Settings → Billing**, add a payment method and a little prepaid credit, and set a **monthly spend limit** — this cap is your real safety net. Then under **API Keys**, create a key (it starts with `sk-ant-` and is shown once).

2. Deploy the Worker. Create a Cloudflare Worker, paste in the included `worker.js`, add your key as the secret `ANTHROPIC_API_KEY`, and set the plain variable `ALLOWED_ORIGINS` to your site's origin (for example `https://yourname.github.io`). The Worker injects the key, only answers requests from your origin, and caps the model and token count.

3. Point the app at it. In `index.html`, set the `PROXY` constant to your Worker URL. The app then calls the Worker instead of Anthropic directly, so no key is ever in the browser. The included `DEPLOY-PROXY.md` and `QUICKSTART.md` walk through every step.

IMPORTANT — Keep the key secret, and keep the spend cap

The key belongs only in the Worker's secret store — never in index.html or the public repo. Lock ALLOWED_ORIGINS to your own site so others can't borrow your proxy, and keep the monthly spend cap on the Anthropic account: that cap is the one hard limit that cannot be bypassed.

NOTE — How safe is the proxy?

It is “safe enough for family,” not Fort Knox. It is origin-locked and model/token-capped, which stops casual misuse, but a determined person who knows the URL could still send some traffic. The spending cap is the backstop that keeps any abuse cheap.

TIP — Roughly what it costs

The dashboard uses Claude Sonnet for short, infrequent calls (a prediction, a bio, a clip lookup) — typically a few cents a day across a small group. Your spend cap is the hard ceiling.

CHAPTER 9

Troubleshooting & FAQ

I updated the app but my phone still shows the old version.

Phones and installed web apps cache aggressively. Hard-refresh the page, or if you added it to your home screen, remove that icon and re-add it after reloading.

A goal clip says “no official clip found.”

The official video for that exact goal wasn't found on YouTube, or the AI couldn't resolve one. Tap to retry. Most goals resolve; some genuinely have no embeddable clip.

The bell / alerts don't seem to do anything on my phone.

Browser goal alerts only fire reliably on desktop with the tab in the background. On phones the operating system suspends the page's timers in the background, so it can't detect a goal to notify you. While the app is open you still get the in-app goal pop-ups and celebration.

Predictions or bios won't load.

In this build they run through the proxy, so there is nothing to enter. If they fail, the proxy or its underlying key/billing may need attention (Chapter 8) — the core scores and tracker keep working regardless.

The live pitch ball sits in midfield.

It leans into the dominant team's half and reacts to commentary events; in a genuinely even, quiet passage it spends time near the center. It is an approximation of play, not real ball tracking.

Do I need to be online?

Yes for live data. The app caches the last data it loaded so it opens instantly and can show the most recent state briefly, but live updates need a connection.

A player shows a flag instead of a photo.

Photos come from ESPN's athlete data and are cached. Where ESPN has no headshot on file for that player, the app shows the team flag instead — that is the intended fallback, not an error.

How do I see what is going wrong under the hood?

Add `?debug=1` to the address, or tap the version tag in the footer five times. A hidden diagnostics panel then lists any malformed data or errors with timestamps and a Copy button. It is invisible to everyone else.

CHAPTER 10

Technical Deep Dive (Developers)

This chapter is for anyone who wants to understand, modify, or re-deploy the app.

10.1 What it is

A single self-contained `index.html` — roughly 2,900 lines / ~190 KB: a hand-written CSS block plus a large JSX/JavaScript section and the HTML head. It is a client-only React application with no backend, no database, and no build step.

10.2 Stack & tooling

UI	React 18.3.1 (UMD) + ReactDOM, rendered with hooks.
Transpile	Babel-standalone 7.28.4 compiles the JSX in the browser at load time (a <code><script type="text/babel"></code> block).
CDN	All libraries load from <code>cdnjs.cloudflare.com</code> (the only allowed external script host).
Styling	Hand-written CSS with CSS custom properties for theming. Fonts: Anton + Inter (Google Fonts).
Build	None. No bundler, no TypeScript, no package step — edit the file and reload.
Hosting	GitHub Pages. The file must be named <code>index.html</code> at the repo root to serve as the site homepage.

NOTE — Why no build step

The no-tooling design is deliberate: the whole app is one file you can edit in any text editor and deploy by dragging it into a repo. The trade-off is in-browser Babel compile cost at load and no type-checking — acceptable for a personal project.

10.3 Data sources

High-frequency data comes from ESPN's free public soccer API (no key, CORS-open). Base path:

```
https://site.api.espn.com/apis/site/v2/sports/soccer/fifa.world
```


/scoreboard	All fixtures, scores, status, events (goals/cards), and team stats in one call. Parsed by <code>parseEvent()</code> .
/summary?event={id}	Per-match ball-by-ball commentary and lineups, parsed by <code>espnSummary()</code> . Drives the live commentary feed and the pitch ball.
/news	Headlines for the News tab.

Everything else is **derived on the client** from the scoreboard: standings (`deriveStandings`), top scorers (`deriveScorers`), a player's goals (`playerGoals`), the knockout tree (`buildBracket` + `resolveBracket`), qualification scenarios (`roadToR32`, which brute-forces every remaining group result via `ggEnum`), the cross-tournament goal ticker (`liveTicker`), a team's projected knockout route (`pathToFinal` + `projWinner`), live win probability (`winProb`), momentum, and team colors (`teamColorMap`).

10.4 The AI layer

All AI calls go through one helper, `callClaude(system, prompt, maxTokens)`. When the `PROXY` constant is set (the default in this build), it posts to the Cloudflare Worker with **no key in the browser** — the Worker injects the key, verifies the request origin, and caps the model and tokens:

```
fetch(PROXY, {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: { model: "claude-sonnet-4-6", max_tokens, system, messages }
})
// Worker adds x-api-key + anthropic-version server-side,
// checks Origin against ALLOWED_ORIGINS, forwards to Anthropic.
```

If `PROXY` is empty (a self-host build with no proxy), the same helper falls back to calling `api.anthropic.com` directly with a user-supplied key from `localStorage` and the `anthropic-dangerous-direct-browser-access` header; in that mode a Settings gear is shown so each visitor can add their own key.

Either path is used by `fetchAnalysis` (predictions), `fetchBio` (comprehensive player profiles), `fetchNewsLLM` (news fallback), `fetchClip` (resolving an embeddable YouTube highlight + timestamp) and `fetchFun` (curating the Fun-tab reel). All use the web-search tool; responses are parsed defensively with an `extractJSON` helper.

10.5 Component map

App	State hub: holds board data, theme, favorites, the open modal, polling, and renders the tabs.
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MatchCard	A single fixture card on the Matches tab.
MatchModal	The live tracker / review / preview. The largest component.
PitchView	The animated pitch; positions the ball from commentary + momentum.
LiveStat / Spark	Animated stat tiles and the momentum sparkline.
PlayerModal / GoalClip	A player's goals + auto-loaded full profile; GoalClip is the in-app trimmed highlight with block-detection fallback and per-goal caching.
TeamModal	A nation's page: stats, form, star players, the full squad (assembled from recent line-ups), tappable recent results that open the match review, and "Make my team."
RoadCard	Renders a group's Road to the Round of 32 scenario card from roadToR32 output.
LiveTicker	The tournament-wide live score/goal ticker beneath the tabs; flashes when a new goal lands anywhere.
PathToFinal	A favorite team's projected route through the bracket to the Final.
Headshot	Resolves a player's photo from the ESPN athlete endpoint (cached) with a flag fallback (10.9).
Boundary / DebugPanel	The error boundary and the hidden diagnostics panel (10.10).
FunTab / VideoCard / FunModal	The Fun reel: filter chips (including a viral Most Viewed cut); thumbnails open FunModal, an expanded lightbox player with the watchdog fallback and tap-for-sound.
AlertBell / Settings	Goal-alert permission and the API-key field (hidden under a shared proxy).
Mark / Star / StatBar / Countdown	Small shared UI primitives.

10.6 Theming engine

Picking a favorite calls `setFavoriteTeam`, which writes the team's ESPN colors into CSS variables `--teamA` / `--teamB` on the document root. `bright()` lightens a too-dark

secondary color so accent text stays legible; `ink()` picks black-or-white for contrast on filled buttons. The group label deliberately uses the fixed gold accent so it's readable under any theme.

10.7 Live-data flow & polling

- Board (scoreboard) polls every **12s** when any game is live, otherwise every **120s**.
- An open live tracker refreshes its match every **~5s** for near-real-time updates; new goals append incrementally without reloading existing clips.
- A 1-second ticker drives the match clock and countdowns.
- All polling pauses when the tab is hidden; the last board is cached in `localStorage` for instant load.

10.8 Caching & preloading architecture

The app pre-warms data aggressively so taps feel instant and AI cost stays low. Four caches persist in `localStorage` (all namespaced under `wc2026_`):

board	The last scoreboard, for an instant cold-start render.
bio_{name+team}	A player's full AI profile. 7-day TTL (<code>bioCacheGet</code> / <code>bioCacheSet</code>).
squad_{team}	A team's roster, assembled from recent line-ups. 7-day TTL (<code>fetchSquadFor</code>).
hs_{id}	A resolved player-photo URL. 30-day TTL, plus an in-memory <code>hsMem</code> map (see 10.9).

Two background warmers fill the bio cache:

- **Golden Boot warmer** — on each board load it walks every top scorer (`deriveScorers`), warms their bio (throttled `~450ms`), and pre-resolves their photo.
- **Round-of-32 warmer** — walks your favorite team, then the top `PRELOAD_TOPN` (default 2) of every group; for each it loads the squad (cached) then warms each player's bio, starters first, throttled `~800ms`. It keys off a stable signal so it does not restart on every score poll.

`warmBio` is a shared wrapper with an in-flight Set, so the two warmers never fetch the same player twice. `PRELOAD_BIOS` is the master on/off switch; `PRELOAD_TOPN` trades coverage against cost.

HONEST LIMIT — Warming costs money the first time

A full warm is hundreds of AI calls. The 7-day cache turns that into a roughly once-a-week cost, and your Anthropic spend cap is the hard ceiling. Lower `PRELOAD_TOPN`, or set `PRELOAD_BIOS = false`, to trim it.

10.9 Player photos — the Headshot pipeline

ESPN's scoreboard rarely includes a player's headshot, so the Headshot component resolves the real photo from the athlete endpoint and caches it:

```
GET .../fifa.world/athletes/{id}
  -> athlete.headshot.href    (parsed defensively across shapes)
```

Resolution is cached 30 days (`hs_{id}`) with an in-memory map and an in-flight de-dupe, so each player is fetched once. Rendering falls back gracefully:

- the explicit headshot from the feed, else
- the resolved athlete-endpoint URL, else
- a direct ESPN CDN guess (the image `onError` catches a miss), else
- the team flag (Mark).

Photos appear in the Golden Boot list, the player-profile header, team-page star chips, and every goal row; the Golden Boot list pre-resolves its photos on board load.

10.10 Resilience: diagnostics, `safe()`, and the error boundary

Because external feeds can change shape, several layers keep a bad payload from blanking the app:

- **`safe(fn, fallback, label)`** wraps every render-time derivation (`deriveStandings`, `deriveScorers`, `roadToR32`, the bracket). A throw degrades just that section to a safe fallback and is logged — the render never dies.
- **Hardened parsers.** `espnScoreboard` shape-checks the events array and counts skips; `espnSummary` guards JSON and derivation; `parseEvent` logs skipped events; `extractJSON` logs labeled failures, and each AI fetcher tags its parse (`clip`, `fun`, `mostViewed`, `prediction`, `bio`, `news`).
- **Boundary** is a React error boundary wrapping the whole app and each modal; a render throw shows a friendly fallback instead of a blank screen.
- **DIAG** is a 60-event ring buffer recording malformed payloads and crashes, surfaced by the hidden `DebugPanel`.

TIP — Turn on the diagnostics panel

Append `?debug=1` to the URL, or tap the build tag in the footer five times. A small panel lists every malformed payload or render crash with timestamps and a Copy button — invisible to normal users.

10.11 The qualification engine (Road to R32)

`roadToR32(board, standings)` computes each team's fate per group by brute force:

- `gqEnum(k)` enumerates all 3^k win/draw/loss combinations of the k remaining group games (capped at 6 → at most 729 combinations).

- For each combination it clones the live table, applies results (winner +3 points and a +1 goal-difference approximation), re-sorts by points → GD → GF, and tallies how often each team lands top-2 / top-3 — overall and conditioned on its own next result.
- Status: top-2 in every combination → *through*; never top-3 → *out*; always top-3 → *safe3*; otherwise *alive*, with a plain-English line built from the conditional tallies.

The points maths is exact; goal-difference tiebreakers are approximated, and the eight-best-third race spans all groups, so a *safe3* team is not mathematically guaranteed. `pathToFinal` + `projWinner` reuse the same standings strength to project a team's knockout route.

10.12 Editing the file

Open `index.html` in any editor; the app is the JSX inside the `<script type="text/babel">` block. Common edits:

Change the AI model	Search for <code>claude-sonnet-4-6</code> and replace the model string.
Set / change the AI proxy	Edit the <code>PROXY</code> constant to your Cloudflare Worker URL (empty string falls back to a per-user key in Settings).
Adjust polling speed	Search the <code>setInterval</code> values (12000 / 120000 / 5000 / 1000).
Re-tune the live pitch ball	Edit <code>inferTarget</code> (event keywords → pitch zones) and the flow interval in <code>PitchView</code> .
Change the win-probability model	Edit the <code>winProb</code> function.
Restyle	Edit the CSS in the <code><style></code> block, or the CSS variables at <code>:root</code> for the base palette.
Storage namespace	All <code>localStorage</code> keys are namespaced (<code>wc2026_...</code>); bump it to reset persisted state.

TIP — Validate before you deploy

Because there's no build step, a JSX typo only shows up at runtime (a blank page). Test by opening the file in a browser and checking the console before pushing. Keep a last-known-good copy so you can revert rather than layering fixes.

10.13 Build & validation workflow

There is no build step, so a JSX typo only surfaces as a blank page at runtime. Two quick checks catch most issues before deploy:

- Extract the babel `<script>` block and run it through **esbuild** (a fast parse) to catch syntax errors.
- Optionally run a TypeScript checkJs pass (`allowJs`, `noUnusedLocals`) to flag undefined names and dead variables — the code is plain JS, but this surfaces real mistakes.

Bump the BUILD constant on every deploy; it renders in the footer (e.g. v40) so you can confirm which version actually loaded past the cache. Keep a last-known-good copy to revert to rather than layering fixes.

10.14 Deploying to GitHub Pages

- Put `index.html` at the repository root.
- In the repo: **Settings** → **Pages**, set the source to your main branch.
- Your site serves at `https://<user>.github.io/<repo>/`.
- After each push, hard-refresh (or remove/re-add the home-screen icon) to clear the cache.

10.15 Known constraints

- **No isolated goal video.** Clips are official YouTube highlights, trimmed/cued — not rights-cleared broadcast cuts.
- **No real ball tracking.** The pitch ball is an event/possession-driven approximation; live coordinates aren't in any free feed.
- **Models, not oracles.** Win probability and momentum are heuristics, labeled as such.
- **Background alerts are limited** on mobile (no service worker / push); foreground in-app alerts work.
- **AI needs the user's key** when self-hosted; ESPN data does not.
- **Qualification scenarios are a projection.** The points maths is exact, but the best-third race and goal-difference tiebreakers are simplified.
- **Path to the Final assumes wins.** Projected opponents use the stronger current seed; it's a route visual, not a forecast.
- **Player photos depend on ESPN.** Where the athlete endpoint has no headshot, the team flag is shown instead.
- **Pre-warming has a first-run cost.** Hundreds of AI calls fill the bio cache once per week; the spend cap governs it.

Appendix

Glossary

Board	The in-app array of parsed matches from the ESPN scoreboard.
Favorite / themed team	The one nation that colors the app and drives the countdown.
Following	Teams starred for the “Following only” filter (can be more than one).
Momentum	A derived 0–100 indicator of which side is on top, from possession and shots.
Win probability	A heuristic home/draw/away estimate from score, time and momentum.
Road to the Round of 32	Per-group scenario card: who's through, alive, or out, and what each team needs.
Best-third race	The contest among third-placed teams for the eight extra knockout berths.
Path to the Final	A favorite team's projected knockout route, opponent by opponent.
Live ticker	The tournament-wide strip of live scores and goals shown while matches are in play.
Player photo (Headshot)	A player's portrait, resolved from ESPN's athlete endpoint and cached, with a flag fallback.
Diagnostics panel	A hidden developer panel (?debug=1, or tap the build tag 5×) listing malformed payloads and render crashes.
Warmer	Background routine that pre-loads bios/squads/photos so taps need no live lookup.

Key endpoints

```
GET /scoreboard?dates=YYYYMMDD-YYYYMMDD # all fixtures + live
GET /summary?event={id}                  # commentary + lineups
GET /news                                  # headlines
```

Anthropic models referenced

In-app default	claude-sonnet-4-6 — fast, low-cost, used for all AI calls.
Access	No user key — AI runs through a Cloudflare Worker proxy that holds the key server-side and is origin-locked and token-capped.
Self-host	Create a key (sk-ant-...) with a monthly spend cap, store it as the Worker secret ANTHROPIC_API_KEY; see Chapter 8.
Console	console.anthropic.com → Settings → Billing, then API Keys.

Constants you can edit (top of index.html)

BUILD	Footer version tag; bump on each deploy.
MODEL	The Anthropic model string (claude-sonnet-4-6).
PROXY	Cloudflare Worker URL; empty string falls back to a per-user key.
PRELOAD_BIOS	Master switch for background bio/photo warming.
PRELOAD_TOPN	Teams per group to pre-warm (default 2 = the auto-qualifiers).
NS	localStorage namespace; change it to reset all persisted state.

localStorage keys (all prefixed with the NS namespace)

board	Last scoreboard for instant load.
theme / favs	Your themed team and your followed teams.
funclips	The current Fun-tab reel (merged seeds + AI picks).
debug	1 when the diagnostics panel is enabled.
bio_* / squad_*	Cached player profiles (7d) and team rosters (7d).
hs_* / clip_*	Cached player-photo URLs (30d) and resolved goal clips.

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